

Homework (Carolina Reaper)

- Q1.** A rhythm pattern has a slope of $m = 3$. What is the slope of a perpendicular line?
(A) $m = -\frac{1}{3}$
(B) $m = 3$
(C) $m = \frac{1}{3}$
(D) $m = -3$
(E) $m = 0$
- Q2.** A canvas size is represented by the equation $y = 2x + 1$. What is the equation of a line perpendicular to it?
(A) $y = -\frac{1}{2}x + 2$
(B) $y = \frac{1}{2}x - 1$
(C) $y = -\frac{1}{2}x - 3$
(D) $y = 2x - 1$
(E) $y = -2x + 1$
- Q3.** The frequency of a sound wave is given by $y = -\frac{3}{4}x + 5$. What is the slope of a line perpendicular to it?
(A) $m = \frac{4}{3}$
(B) $m = -\frac{4}{3}$
(C) $m = \frac{3}{4}$
(D) $m = -\frac{3}{4}$
(E) $m = 0$
- Q4.** A music sheet has two lines represented by the equations $y = x + 2$ and $y = -x - 1$. Are these lines perpendicular?
(A) Yes
(B) No
(C) Maybe
(D) Not enough information
(E) None of the above
- Q5.** A painting has a frame with a slope of $m = -2$. What is the slope of a line perpendicular to the frame?
(A) $m = \frac{1}{2}$
(B) $m = -\frac{1}{2}$
(C) $m = 2$
(D) $m = -2$
(E) $m = 0$
- Q6.** A sound engineer is using the equation $y = \frac{2}{3}x - 4$ to adjust the sound frequency. What is the equation of a line perpendicular to it?
(A) $y = -\frac{3}{2}x + 1$
(B) $y = \frac{3}{2}x - 1$
(C) $y = -\frac{3}{2}x - 2$
(D) $y = \frac{2}{3}x + 4$
(E) $y = -\frac{2}{3}x - 4$
- Q7.** The size of a musical instrument is given by $y = -\frac{1}{5}x + 3$. What is the slope of a line perpendicular to it?
(A) $m = 5$
(B) $m = -5$
(C) $m = \frac{1}{5}$
(D) $m = -\frac{1}{5}$
(E) $m = 0$
- Q8.** A rhythm pattern has two lines with slopes $m_1 = 4$ and $m_2 = -\frac{1}{4}$. Are these lines perpendicular?
(A) Yes
(B) No
(C) Maybe
(D) Not enough information
(E) None of the above

Open Ended Questions (FRQ)

- Q9.** A sound wave has a frequency given by $y = 2x - 3$. Find the equation of a line perpendicular to it that passes through the point $(1, 2)$. Show your work and provide the final equation.
- Q10.** A painting has a frame with a slope of $m = -3$. Find the equation of a line perpendicular to the frame that passes through the point $(-2, 5)$. Show your work and provide the final equation.

Chef's Tips

- Tip 1: Ensure students show their work for free response questions
- Tip 2: Use online resources for additional practice problems
- Tip 3: Encourage students to use graphing calculators to visualize perpendicular lines